

MAJOR PROJECT PROPOSAL

TEAM ID: - 21

PROVISIONAL TITLE: AI + LLM-Driven Early Earthquake and Landslide Prediction with Real-Time Data Streams

SUPERVISOR NAME: - Dr. Silky Sachar

Problem Statement

Natural disasters such as earthquakes and landslides strike suddenly and cause massive loss of life and property. Traditional early-warning systems rely on expensive sensor networks and are often unavailable in smaller cities and rural regions. Even when alerts exist, they are too technical for local decision-makers and citizens, causing delayed responses.

There is a need for a low-cost, AI-based software system that:

1. Uses public real-time data streams (earthquake catalogs, rainfall, weather APIs) combined with synthetic sensor simulations.
2. Predicts earthquake aftershocks and landslide risk using time-series deep learning models.
3. Converts technical risk predictions into human-readable, actionable incident briefs using LLMs and agentic AI.
4. Provides a web-based dashboard for monitoring and auto-generates alerts (SMS/email/WhatsApp).

Tech Stack

- Data & Ingestion: USGS Earthquake API, NASA Global Landslide Catalog, OpenWeatherMap API, Python (requests, APScheduler, Pandas)
- Machine Learning / Prediction: PyTorch (LSTM, Transformer, Autoencoder), Scikit-learn, Tsfresh / PyWavelets
- GenAI & Agents: OpenAI GPT / Hugging Face Transformers, FAISS, LangChain (optional)
- Backend & APIs: FastAPI
- Frontend & Visualization: Streamlit or Next.js + Tailwind, Plotly / Grafana
- Alerting: Twilio (SMS/WhatsApp API) or Email (SMTP)
- Database: PostgreSQL / SQLite, InfluxDB / TimescaleDB (optional)

Work Distribution (2 People)

Chitrabhanu Srivastava– ML + Data Pipeline Engineer

Responsibilities:

1. Data Collection & Preprocessing

- Connect to real-time APIs (USGS, NASA, OpenWeather).
- Build synthetic sensor simulators.
- Clean, normalize, and prepare datasets.

2. Machine Learning Models

- Implement Autoencoder/VAE for anomaly detection.
- Train LSTM/Transformer/TCN for time-series prediction.
- Design ensemble risk-scoring logic.

3. Model Deployment

- Wrap models with FastAPI endpoints for inference.
- Optimize for latency & reliability.

Aamya Pandey – LLM + Dashboard Engineer

Responsibilities:

1. LLM Agent Layer

- Integrate GPT/Hugging Face models to build the Incident Assistant.
- Implement RAG with FAISS.
- Generate human-readable summaries & recommended actions.
- Build agentic automation (auto-generate SMS/email alerts).

2. Dashboard & Visualization

- Create Streamlit or Next.js dashboard with live feeds.
- Risk heatmaps, time-series plots, alert logs with LLM summaries.
- Chat interface with the AI Assistant.

3. Alerting & User Interaction

- Integrate Twilio (SMS/WhatsApp) or email APIs.
- Ensure smooth end-to-end flow (model → LLM → user).

SIGNATURE SUPERVISOR: -

TEAM MEMBERS SIGNATURE: -

Chitrabhanu Srivastava

Aamya Pandey